ANTARA THAPLIYAL

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PROFESSIONAL SUMMARY

Master's student in Computer Science at Rutgers University with 2 years of production software engineering experience. Skilled in **Python, LangChain, LangGraph, MCP, prompt engineering, and GenAI evaluation**, with hands-on experience building AI Agents, agentic workflows, and RAG-based applications integrating vector stores, APIs, and LLM frameworks. Currently developing a multi-agent AI system with real-time tool connectivity and evaluation pipelines. Seeking to apply generative AI engineering and scalable evaluation skills in a fast-paced, innovation-driven environment.

TECHNICAL SKILLS

Languages & Databases: Python, SQL/MySQL, HTML, CSS
GenAI & Agent Frameworks: LangChain, LangGraph, MCP (Model Context Protocol), OpenAI API, CrewAI, FAISS, RAG Architectures
ML Frameworks: PyTorch, TensorFlow, Pandas, NumPy
Core Skills: AI Agent Development, Prompt Engineering & Robustness Testing, Agentic Workflow Design, GenAI Evaluation & Benchmarking, AI Guardrails & Safety, Explainability & Interpretability, Vector Stores/RAG, REST API Integration, Data Pipeline Design, Technical Documentation
Tools & Platforms: Git/GitHub, Jupyter Notebook, Streamlit, VS Code, PyCharm, Microsoft Office
Methodologies: Agile/Scrum, SDLC, Iterative Experimentation & Design, Debugging & Performance Tuning

PROFESSIONAL EXPERIENCE

Tata Consultancy Services (TCS), Pune, India *Jul 2021 – Jun 2023*
Software Engineer – Insurance Domain

- Partnered with cross-functional teams to design and deliver new product features for an enterprise platform, driving a **29% lift in sales**.
- Engineered automated batch-processing data pipelines and letter-generation utilities, cutting manual effort by **30%** and accelerating offline processing workflows.
- Built the "Autopay Re-collection" system integrating database queries and REST API calls for automated loan repayment processing, improving collection efficiency.
- Maintained clear technical documentation, Git-based versioning workflows, and sprint documentation to ensure reproducibility and clarity across engineering teams.

PROJECTS

Shadow Board – Multi-Agent AI Decision System | *Python, LangChain, LangGraph, CrewAI, MCP, REST APIs* *2026*

- Architecting a multi-agent AI system with specialized AI Agents (CFO, CMO, Legal, Devil's Advocate) that independently research via API-based tool integrations and engage in structured real-time agentic workflows with evaluation checkpoints.
- Engineered prompt engineering strategies and prompt robustness testing for distinct agent personas, optimizing response quality, safety guardrails, and inter-agent collaboration using LangChain and LangGraph orchestration.
- Designed evaluation harnesses and benchmarking pipelines to assess agent decision quality across diverse scenarios, iterating on agent behaviors through systematic experimentation and documentation.

Capital Finder Bot – RAG-Based AI Chatbot | *LangChain, OpenAI API, FAISS, Streamlit, Python* *2025*

- Developed a RAG-powered chatbot integrating LangChain, OpenAI GPT-3.5 API, and FAISS vector store for context-aware retrieval; applied prompt engineering and token optimization to reduce response latency by **78%**.
- Documented engineering workflows, API integration patterns, and evaluation results to ensure reproducibility and support iterative design improvements.

Beside Her – AI-Powered Health Platform | *Flask, Gemini API, ML Pipeline, REST APIs* *2026*

- Built a full-stack AI application with a 7-technique ML pipeline (clustering, anomaly detection, trend analysis) integrated with Google Gemini API for personalized guidance through REST API endpoints and scalable data pipelines.

Incentive-Compatible Reward Design for Adaptive AI Agents | *Python, Q-Learning, NumPy* *2025*

- Conducted evaluation and benchmarking experiments across 4 reward schemes, training agents under non-stationary conditions with concept drift, demonstrating adaptive recovery and presenting comparative findings on calibration accuracy.

EDUCATION

Rutgers University, New Brunswick, NJ *Expected Dec 2026*
Master of Science in Computer Science | GPA: 4.0

G.B. Pant University of Agriculture & Technology, India *Jun 2021*
Bachelor of Technology in Information Technology

LEADERSHIP & ACTIVITIES

- **Astitva NGO (2017–2021):** Dedicated three days weekly to teaching underprivileged children, enhancing educational access and community wellbeing.
- **Event Management:** Coordinated a 25th Alumni Reunion and organized Photoholic, awarded Best Event for two consecutive years.
- **Certification:** Supervised Machine Learning: Regression and Classification – Stanford Online & DeepLearning.AI (Coursera)